

Introduction

This lab was very similar to the previous Lab 5, so I used a save as on the previous lab to make my new one. Recall, we are using a rectifier circuit with a noisy input signal expected to output a waveform that is always positive. Rectifier circuits are useful in real-world applications for converting between AC and DC power.

The only thing that changed was the use of a while loop instead of a for loop, the use of a timer to be able to use time as a parameter, and the use of MATLAB to analyze the code. I also had to adjust the function generator simulating the signal to simulate acquisition timing to get values the same as the timer parameter.

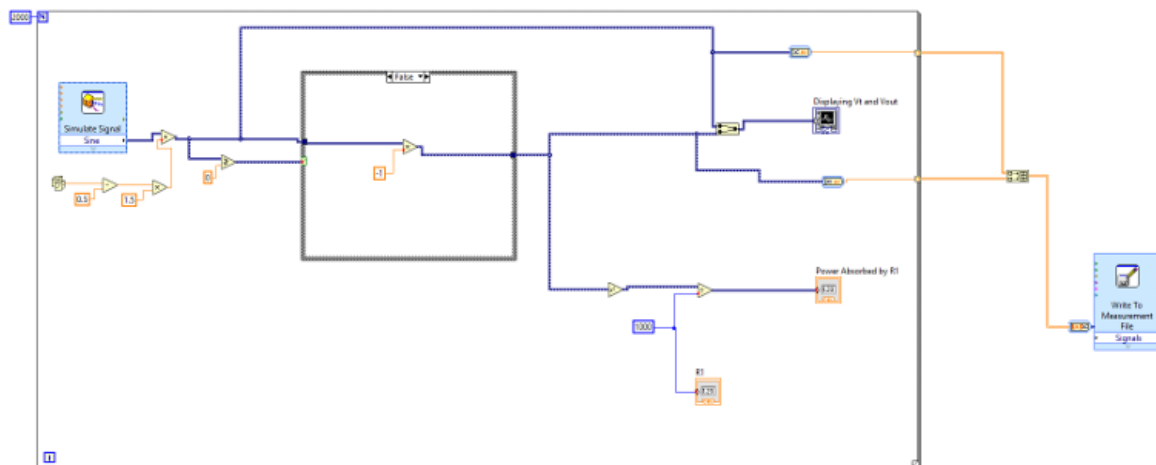
Lab 5 Front Panel:

Front Panel:



Lab 5 Block Diagram:

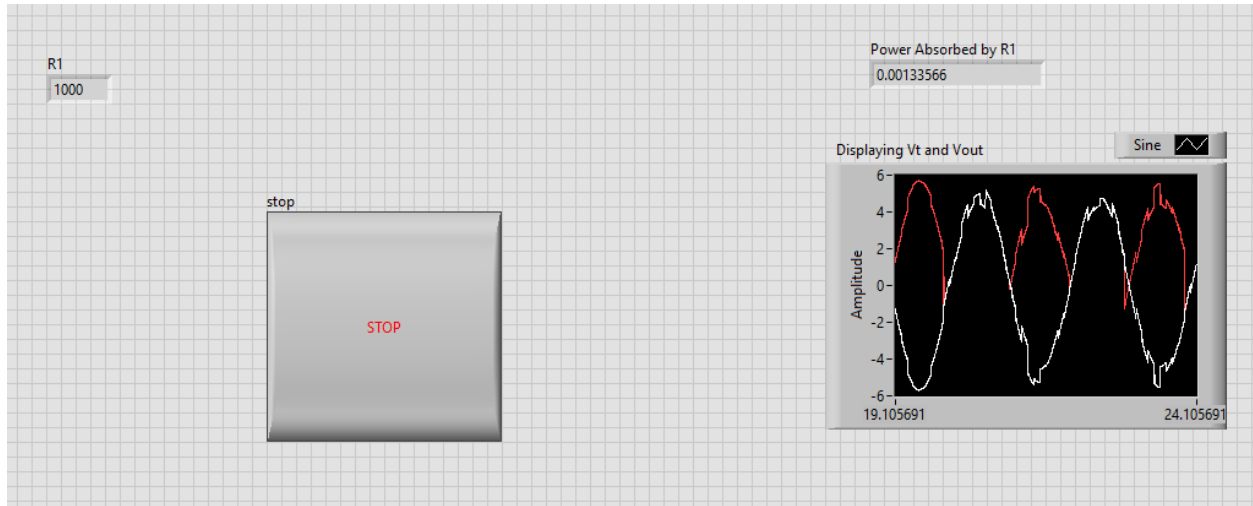
Block Diagram:



Analysis

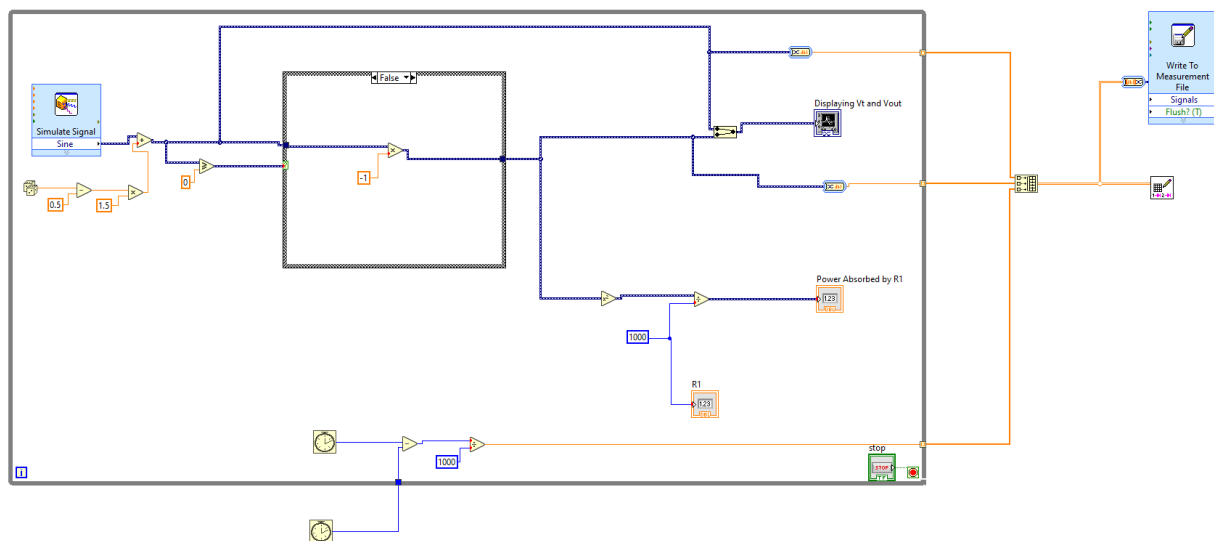
After all my adjustments were made this is what my new LabVIEW looked like:

LabVIEW Front Panel:

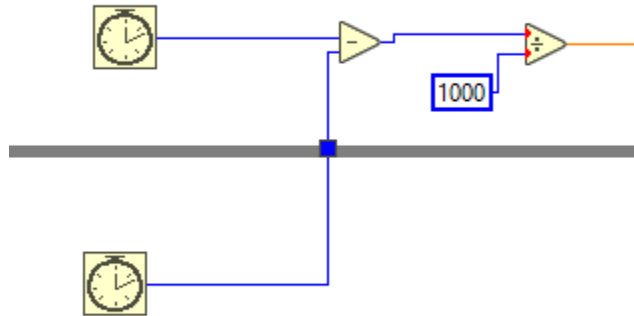


Note: the time goes to about 24 seconds and this matches with the Excel and MATLAB code

LabVIEW Block Diagram:

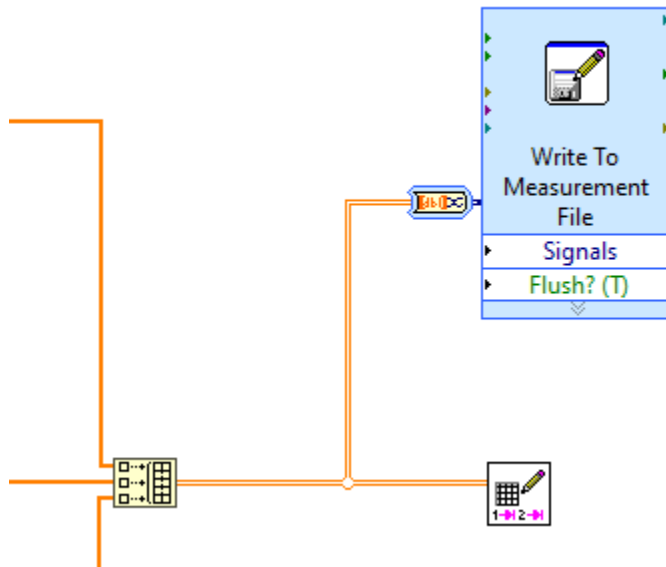


Time parameter set up:



I made two timers, one inside the loop and one outside. I subtracted them from each other to initialize the timer inside the loop at 0. Then I divided by 1000 to convert from milliseconds to seconds. Then I expanded the array and connected the timers to it. Notably since I connected it to the new, third part of the array my time appears as the third row (columns in excel) and I adjusted my MATLAB code accordingly.

Writing to Excel file and Delaminated Spreadsheet:



I knew I needed to put the data into an Excel spreadsheet, so I used the Write to Measurement (MEAS) file tool like I did in Lab 5 making a few adjustments like file name. This was useful for later analyzing and verifying I got the right data.

I was unsure how to use an excel file with MATLAB, so I simply added an additional output to my array to use the Write Delaminated Spreadsheet tool. Then I was able to use this generated file with MATLAB to further analyze the data and plot the waveforms again.

Sample of Excel file:

	A	B	C
1	Vt	Vout	Time
2	2.045387	2.045387	0
3	2.523904	2.523904	0.101
4	4.334079	4.334079	0.2
5	5.442453	5.442453	0.301
6	5.053553	5.053553	0.401
7	5.28131	5.28131	0.501
8	4.214659	4.214659	0.6
9	3.247969	3.247969	0.7
10	1.207664	1.207664	0.801
11	0.479185	0.479185	0.9
12	-0.84056	0.840556	1
13	-2.27357	2.273566	1.101
14	-4.11749	4.117489	1.2
15	-5.46994	5.469941	1.301
16	-4.85225	4.852253	1.4

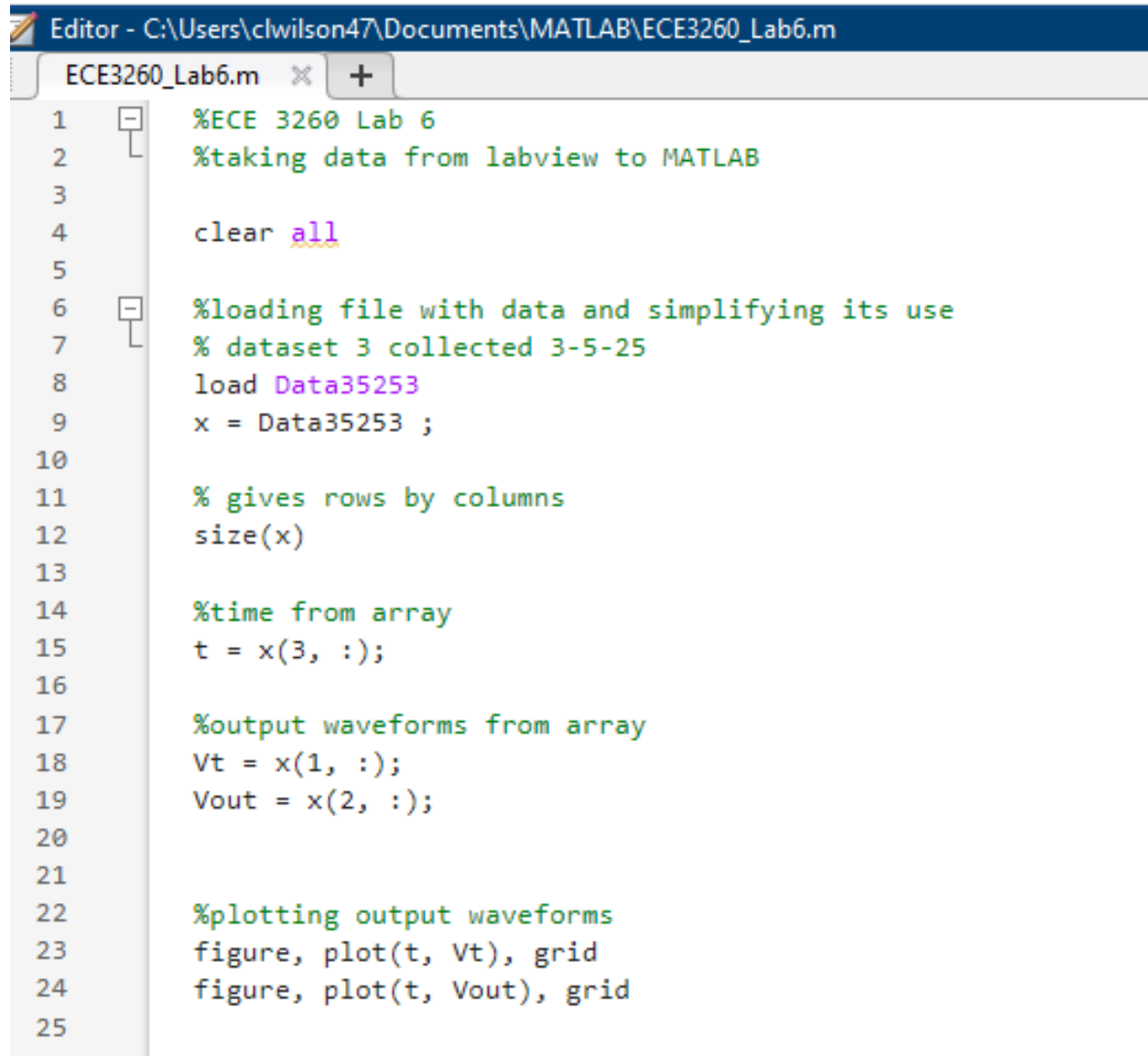
Note: the time goes to about 24 seconds as expected from the LabVIEW waveform chart

241	0.050499	0.050499	23.9
242	1.155709	1.155709	24
243			

The Excel file is useful for checking and verifying which parameter is which, and if you wanted to do more data analysis using Excel (mean, median, probabilities, etc.) this may be helpful.

MATLAB code:

I wrote a simple script to plot the values of the waveforms versus time to display the waveforms again as MATLAB figures.

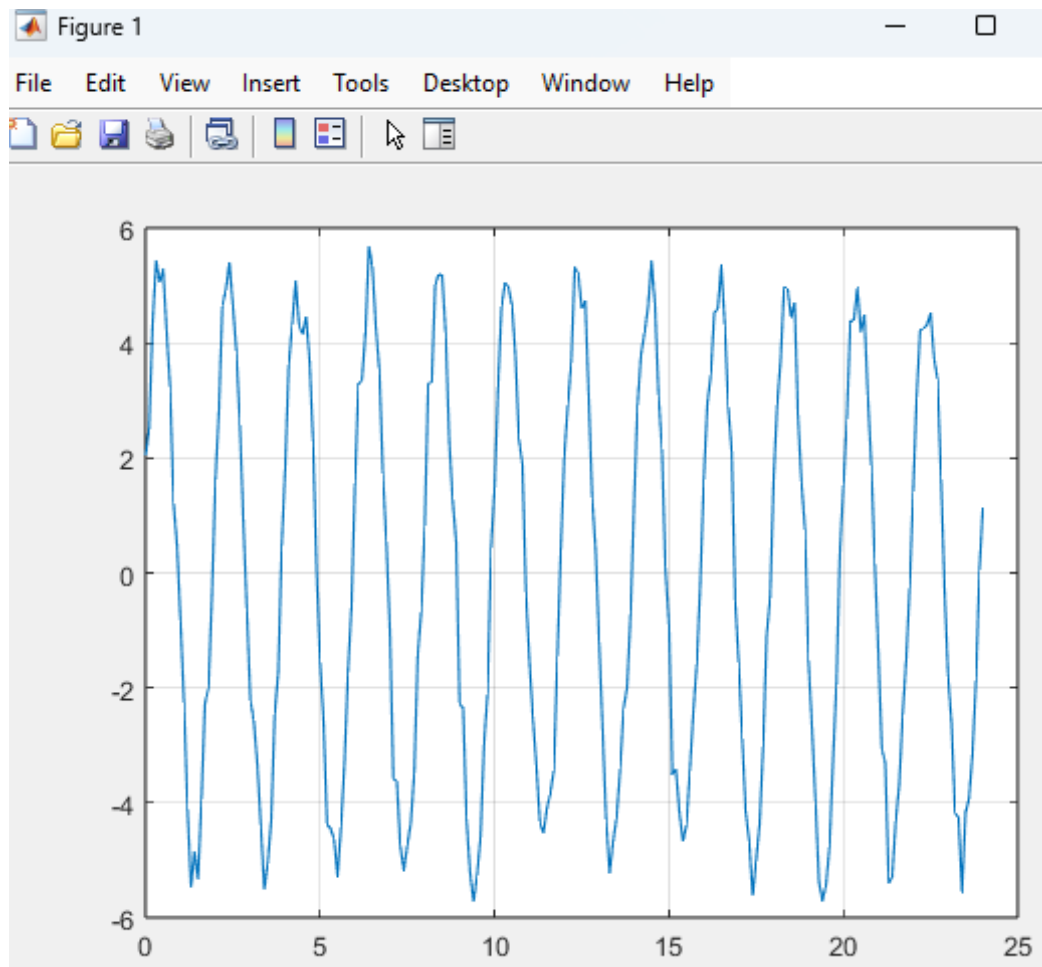
The image shows a screenshot of the MATLAB Editor window. The title bar at the top reads "Editor - C:\Users\clwilson47\Documents\MATLAB\ECE3260_Lab6.m". Below the title bar, there is a tab labeled "ECE3260_Lab6.m" with a close button (X) and a plus sign (+). The main area of the window contains a MATLAB script with 25 lines of code. The code is as follows:

```
1 %ECE 3260 Lab 6
2 %taking data from labview to MATLAB
3
4 clear all
5
6 %loading file with data and simplifying its use
7 % dataset 3 collected 3-5-25
8 load Data35253
9 x = Data35253 ;
10
11 % gives rows by columns
12 size(x)
13
14 %time from array
15 t = x(3, :);
16
17 %output waveforms from array
18 Vt = x(1, :);
19 Vout = x(2, :);
20
21
22 %plotting output waveforms
23 figure, plot(t, Vt), grid
24 figure, plot(t, Vout), grid
25
```

Note: I used the Excel spreadsheet and Block diagram to verify which array parameter was which when I was setting t , V_T , and V_{OUT} .

MATLAB figures:

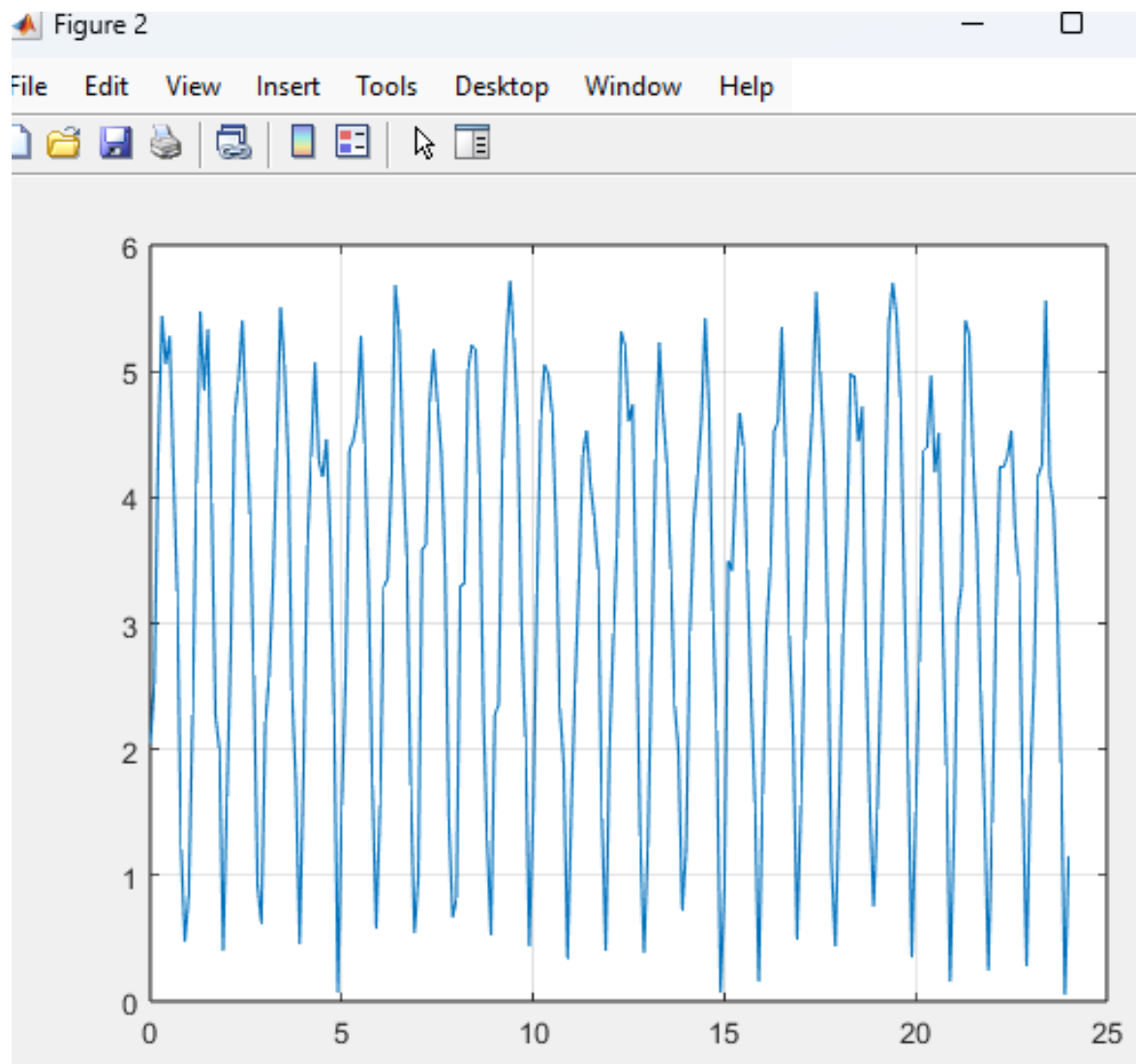
Graph of V_T :



Note: this one has positive and negative values with some visible noise as expected

Note: the time goes to about 24 seconds as expected from LabVIEW and Excel

Graph of V_{OUT} :



Note: this one does not have any negative values as expected due to the rectifier

Note: the time goes to about 24 seconds as expected from the other MATLAB figure, Excel, and LabVIEW

Conclusion

This lab was fun. It was nice to see that the same data generated using LabVIEW could be stored and analyzed so many ways.

We were able to apply what we learned about using MATLAB to a real problem and ensure that the x-axis meant something (time) rather than being a bunch of random numbers.

This will be extremely useful in the future for analyzing real-world systems that almost always depend on time, and being able to export the data to other tools with better data analysis capabilities than LabVIEW.